



Copyright © 2024 International Journal of Cyber Criminology – ISSN: 0974-2891
January – June 2024. Vol. 18(1): 130-147. DOI: 10.5281/zenodo.4766807

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Development and Evaluation on Cybersecurity Behaviour Measurement Instruments for Undergraduate Students

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Abstract

*Technological advancements from past to present have ushered the world into an era of rapid information exchange, establishing a networked system that has transformed information systems and introduced a new realm known as the "cyber world." While this progression has rendered access to information more efficient and convenient, it has concurrently spurred a rise in cyber threats, which have evolved into a societal concern and contributed to other criminal activities. Assessing knowledge for adaptability and promoting cybersecurity awareness is, therefore, an imperative for all sectors. This quantitative research aims to develop and evaluate a cybersecurity behaviour measurement instrument. The sample comprised 820 undergraduate students, with sample size determination conducted via the G*Power programme. The measurement instrument, initially designed by the researchers, encompassed 100 items across four components: 1) Awareness of cyber threats, 2) Knowledge of cyber threats, 3) Experience with cyber threats, and 4) Self-protection against cyber threats. Following expert evaluation, 67 items remained, rated on a 6-point scale. The findings supported*

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all three hypotheses, with 21 items meeting established criteria, enabling the instrument to explain 64.187% of cybersecurity behaviours among undergraduate students and achieving an average reliability score of 0.885. Additionally, the model demonstrated alignment with empirical data, meeting standard indices, including $\chi^2 = 94.392$, $df = 59$, $p\text{-value} = 0.531$, $RMSEA = 0.044$, $CFI = 0.987$, $TLI = 0.986$, and $SRMR = 0.062$. In conclusion, the researchers recommended potential applications for the instrument in developing social indices and proposed extending it to study causal and effect factors in future research.

Keywords: Cybersecurity Behaviours, Measurement, Undergraduate Students.

Introduction

Technological advancements, particularly in communication technology, the internet, and wireless electronic devices, have profoundly impacted daily life. These innovations have enhanced convenience in daily activities, enabling clearer and faster communication, simplified access to research information, and seamless social media connectivity. Information and communication technologies have developed rapidly across borders, allowing universal access and benefits. The foundation of modern communication lies in the internet—a vast global network of interconnected systems that facilitates immediate interactions regardless of time or location. However, this infrastructure has also created opportunities for cybercrime, transforming the internet into a platform for criminal activities. Cybercrime has evolved beyond traditional offences, exhibiting changes in methods, frequency, speed, and scale of impact, with incidents becoming increasingly severe and widespread.

Statistics from the Royal Thai Police on online fraud incidents between January and June 2024 reveal 50,781 cases, with total damages exceeding 40,475 million baht—an average of 224 million baht per day. Among these cases, 19.22% involved individuals aged 18–22, typically late teenagers or undergraduate students (Churprakobkit et al., 2023). These figures highlight the increasing impact of cybercrime on individuals' lives and property, with substantial financial losses. Undergraduate students are particularly vulnerable to online fraud due to two main factors: 1) teenagers and students are frequent internet users; and 2) a study by Watts et al. (2017) indicated that Problematic Internet Use (PIU), characterised by difficulties in controlling internet usage and resulting in stress, is more prevalent among students than other population groups.

As information technology advances swiftly to address user needs, it simultaneously gives rise to cybersecurity threats and attacks on information systems across all organisational tiers. The frequency of cyberattacks is escalating, encompassing activities such as hacking into information systems, data theft, social media account impersonation, identity theft for fraudulent purposes, and a range of other cyber threats. Undergraduate students represent a transitional phase into adulthood, with technology integrated into their daily lives. They quickly adapt to technological advancements, which encompass skills in information access, data retrieval, digital media production, and content sharing (Salubi et al., 2018). However, research by Yan et al. (2018) revealed that only 62% of the 462 surveyed students in the United States considered cybersecurity in their decision-making, highlighting a

concerning gap given the potential consequences. Additionally, statistics indicate that individuals aged 18–24, despite being skilled digital users, are more vulnerable to cyber threats than older age groups, such as those aged 25–40, who tend to be more cautious in managing privacy and security. This heightened awareness among older users stems from greater knowledge and experience in recognising digital risks. Furthermore, engaging in digital transactions increases exposure to cyber threats (Debb et al., 2020). Victims of cyber threats often face financial, emotional, and physical repercussions. Thus, individuals with cybersecurity training or previous experiences are generally more aware of these dangers. Ahmad et al. (2021) emphasised the importance of educating students to establish a foundation for future cybersecurity, enabling them to safeguard themselves against cyber threats.

In addition to existing threats, cyber threats have evolved, with cyber-attacks adapting their methods to exploit new vulnerabilities. These contemporary forms of attacks differ from traditional ones, often surprising internet users and hindering their ability to monitor or protect themselves as effectively as before. This evolution renders users susceptible to unforeseen harm or damage. Awareness of cybersecurity is essential for preparing and strategizing against such threats, enabling better self-protection against various risks. Individuals with adequate knowledge are more likely to recognise and anticipate potential harms. Although cyber threats can impact anyone, they disproportionately target university students compared to other age groups. These students, having grown up alongside technological advancements and being accustomed to navigating the digital landscape, face a heightened risk of becoming victims of cybercrime.

A review of the research literature indicates a significant absence of internationally standardised instruments for measuring cybersecurity behaviour among undergraduate students. Consequently, the researchers are motivated to study and develop an instrument specifically designed to assess cybersecurity behaviour within this demographic. The objective is to create a robust assessment tool that undergoes rigorous validation in accordance with stringent academic standards, enabling a comprehensive evaluation of cybersecurity behaviours. Moreover, this instrument will facilitate the screening of students' cybersecurity behaviours, thereby enhancing individual awareness of cyber threats in the future.

Literature Review

Thailand has fully transitioned into a hyper-connected digital society, where electronic devices are essential for daily life, allowing continuous online engagement. The COVID-19 pandemic has accelerated this shift, establishing a new normal for global society. People have become accustomed to conducting various activities online, including work, education, communication, and financial transactions. A study by Bicen et al. (2011) revealed that over 32% of undergraduate students spend more than four hours daily on social media, while Jones et al. (2009) found that 97% of students use the internet for communication. The research also indicated that students seek information related to entertainment, education, and personal matters, with some engaging in illegal activities. Additionally, Sherman et al. (2000) highlighted that technology offers varying benefits based on users' backgrounds and interests; however, hidden dangers exist if users do not exercise adequate caution.

As society adapts to the new normal, many individuals continue to conduct transactions online, leading to an increase in electronic data storage and a heightened risk of data breaches from cyberattacks. These digital threats, which involve unauthorized access to systems and data, can cause significant harm to individuals and organisations (Chaikin, 2006). With the growing volume and variety of online activities, including shopping and cloud data storage, understanding how to protect oneself from cyber threats has become essential (Zwilling, 2021). Cybersecurity awareness is crucial for compliance with security protocols. Research by Duzenci et al. (2023) indicates that individuals' decision-making processes significantly influence their cybersecurity compliance. Similarly, Quayyum et al. (2021) found that cybersecurity knowledge and awareness greatly affect individuals' preventive behaviours against cyber threats. Increased exposure to cybersecurity information leads to better understanding and safer online practices. However, studies have shown that many individuals, particularly students, often lack basic cybersecurity knowledge and neglect necessary precautions. Russo et al. (2023) noted that students using smartphones frequently ignore cybersecurity risks, while Kovačević et al. (2023) highlighted that students typically lack awareness of how to protect themselves. A survey by Djeki et al. (2024) revealed low levels of cybersecurity awareness among students and faculty, indicating that foundational knowledge is a critical factor influencing cybersecurity behaviour (Quayyum et al., 2021).

Cyber threats can be categorised into eight distinct areas, as identified by Frisk et al. (2023). These categories include abusive content, which refers to the use or dissemination of false or inappropriate information aimed at damaging the credibility of individuals or institutions, resulting in unrest or the distribution of illegal content. Availability attacks target system functionality, leading to delays or making systems inoperable. Fraud encompasses attempts to gain advantages through deception or fraudulent practices, which may manifest in various forms, such as unauthorised system usage. Information gathering involves efforts to collect data from systems without proper authorisation. Intrusions denote successful unauthorised access to a system, resulting in control being taken by an unauthorised individual, while intrusion attempts are efforts aimed at breaching a system's security. Malicious code or malware refers to software designed to inflict damage on systems, posing a significant threat to information security. Finally, information security concerns the unauthorised access to or alteration of sensitive information.

The concept of cybersecurity encompasses the strategies and actions designed to prevent, mitigate, and minimise risks associated with cyber threats that may compromise the confidentiality, integrity, and availability of information and equipment within an information system. Inadequate cybersecurity measures can leave users vulnerable to harm and expose their personal data to malicious actors. Enhancing cybersecurity can be achieved through various methods, including the use of complex passwords and the installation of antivirus software (Florackis et al., 2023). This perspective aligns with Maslow's Hierarchy of Needs, particularly the second level, which focuses on the need for safety and security. In the contemporary context, this includes concerns related to internet safety and social media security; individuals who do not perceive themselves as secure are likely to prioritise safety before pursuing higher-level needs for personal development and well-being (Shi et

al., 2021).

The concept of cyber threats encompasses any event or circumstance that poses a risk to the confidentiality, integrity, and availability of information systems or data belonging to individuals, organisations, or nations. These threats primarily stem from cybercriminals or state-sponsored actors (Rahman et al., 2023). Significant categories of cyber threats include ransomware and malware associated with email communications. The concept of cybersecurity is essential for preventing potential threats from cyber disruptions. As the internet is the primary communication channel today, users must adopt strategies to protect against malware, viruses, and other risks that could compromise personal information. Effective cybersecurity requires service providers to implement a comprehensive policy that encompasses network security, including measures to safeguard networks from intruders, maintain application security through regular vulnerability testing and updates, and protect sensitive information by enhancing security measures and restricting access (Azizi et al., 2023).

In the context of individual-level prevention against cyber threats, the protection of personal data and digital assets is crucial to avoid victimisation by cybercrime. Key recommendations for enhancing cybersecurity include: (1) regularly updating software and operating systems, (2) employing antivirus software, (3) creating complex passwords, (4) refraining from opening email attachments from unknown sources, (5) avoiding links from unfamiliar senders or websites, (6) consistently backing up data, and (7) steering clear of insecure Wi-Fi networks, which heighten the risk of cyber-attacks (Safitra et al., 2023).

In the event of a cyber-attack, the following guidelines should be observed: (1) Check for any unidentifiable financial transactions and review credit reports for the emergence of new accounts or loans; (2) Exercise caution against providing sensitive information on websites, via emails, or through social media; (3) If unusual activity is detected, promptly change all account passwords, disconnect all devices, and consult cybersecurity professionals to scan for and eliminate any harmful elements; (4) Report the cyber threat to the appropriate authorities; (5) Conduct a thorough scan of all computer devices to ensure they are free from viruses and operating efficiently; and (6) Disconnect all devices from user accounts and internet networks, and initiate a full system recovery (Darem et al., 2023).

A study conducted by Bidgoli et al. (2016) identified undergraduate students as a high-risk group for cybercrime, primarily due to their pervasive use of technology in daily activities. The research, which surveyed 222 students, found that approximately half had encountered at least one form of cybercrime, with malware, hacking, and scamming being the most prevalent. Additionally, the study indicated that students often learned protective measures and gained cybersecurity knowledge from peers or individuals who had previously been victimized. Complementing these findings, Biese et al. (2024) noted that student victims of cybercrime frequently lack a clear understanding of the extent of the damage and may normalize their experiences due to insufficient knowledge of information networks.

Based on the literature review, a framework for the development and evaluation of measurement instruments is illustrated in Figure 1.

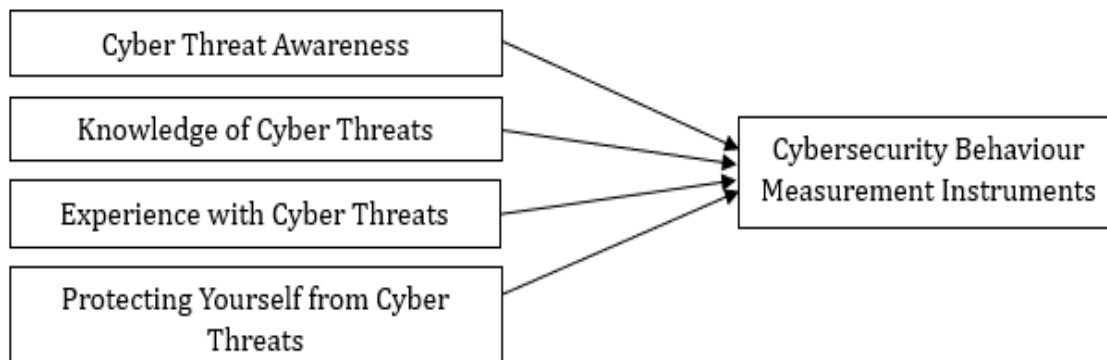


Figure 1: Conceptual Framework.

Methodology

Research Design

This quantitative research study seeks to develop and evaluate a robust measurement instrument for assessing cybersecurity behaviour. The methodology includes item quality testing, exploratory factor analysis, and second-order factor analysis. The results are further validated through confirmatory factor analysis to assess the model's fit with empirical data, employing SEM analysis. This research project has received ethical approval, as indicated by the certification number HREC-130.

Sample

The sample for this research comprises undergraduate students, with the sample size determined using G*Power software. The sampling process involves three stages: 1) Item Quality Testing, utilizing an effect size of 0.75 (medium effect), an alpha level of 0.05, and a power of 0.95 with an allocation ratio of 1. This results in a minimum required sample size of 96 participants based on a t-test for two independent samples; 2) Exploratory Factor Analysis; and 3) Confirmatory Factor Analysis. Following the ten-times rule (Hair Jr et al., 2017; Kock et al., 2018), the necessary sample size is approximately 600 participants. To account for potential incomplete responses, the researchers increased the sample size by 10%.

The researchers employed a multi-stage quota random sampling method to collect data, ensuring equal representation across a large sample (Burger et al., 2006). The process included: 1) random selection of university types, comprising 8 state-supervised and 8 privately supervised universities; 2) selection by faculty groups, including Business Administration, Humanities, Liberal Arts, Engineering, Science and Technology, and Health Sciences; 3) selection of undergraduate students by year—first-year through fourth-year; and 4) division by GPA, categorizing students into low GPAs (≤ 3.00) and high GPAs (> 3.00) using a simple random sampling method. The minimum total sample size was set at 768 participants, with strict controls during data collection to minimize external influences on responses. In this study, data were collected from a total of 833 participants. After filtering for complete responses, the final sample consisted of 820 participants. Table 1 provides preliminary information on the sample utilized for data analysis across the various stages of the research.

Table 1: Preliminary Information of the Samples Used in Various Steps of Analysis.

Preliminary Information of the Samples						
Steps	1 st step (n = 120)		2 nd step (n = 300)		3 rd step (n = 400)	
Analysis	Item Quality Testing		Exploratory Factor Analysis		Confirmatory Factor Analysis	
Samples	Undergraduate Students		Undergraduate Students		Undergraduate Students	
Gender	Male = 58 (48.33%)	Female = 62 (51.67%)	Male = 134 (44.67%)	Female = 166 (55.33%)	Male = 187 (46.75%)	Female = 213 (53.25%)
Average Age (Years)	20.72		20.95		20.81	
GPAX	3.09, SD = 0.21		3.14, SD = 0.27		3.10, SD = 0.23	
Year	1 st Year = 12	1 st Year = 16	1 st Year = 29	1 st Year = 30	1 st Year = 40	1 st Year = 48
	2 nd Year = 14	2 nd Year = 17	2 nd Year = 35	2 nd Year = 42	2 nd Year = 45	2 nd Year = 55
	3 rd Year = 17	3 rd Year = 16	3 rd Year = 37	3 rd Year = 46	3 rd Year = 54	3 rd Year = 61
	4 th Year = 15	4 th Year = 13	4 th Year = 33	4 th Year = 48	4 th Year = 48	4 th Year = 49
Sibling Status	Only Child = 52		Only Child = 138		Only Child = 176	
	Siblings = 68		Siblings = 162		Siblings = 224	
Accommodation	Living in a Dormitory Alone = 30		Living in a Dormitory Alone = 80		Living in a Dormitory Alone = 115	
	Living with Family = 42		Living with Family = 96		Living with Family = 108	
	Living with Friends = 22		Living with Friends = 35		Living with Friends = 76	
	Living with Relatives = 26		Living with Relatives = 89		Living with Relatives = 101	

Note: *Missing Value is Excluded.

Instruments

The development of measurement instruments is a vital aspect of knowledge advancement, as it enables researchers to gain a comprehensive understanding and analysis of individual behaviours. The significance of creating such instruments encompasses three essential components:

- 1) Validity and reliability ensure that the results derived from the measurement instruments are accurate and applicable in both research and practical contexts.
- 2) Standardization facilitates the comparison of results across different sample groups, promoting consistency and broader acceptance in research, thereby making them relevant to specific target populations.

- 3) Reflecting reality means that the instruments must accurately represent the real behaviours being examined, taking into account the evolving social and cultural contexts (Kimberlin et al., 2008).

The process of developing measurement instruments involves seven critical steps, as illustrated in Figure 2.

- 1) The initial step entails a comprehensive review of relevant documents, theories, and prior research to define the variables, formulate hypotheses, and identify the components of the measurement instruments.
- 2) The next step involves constructing the instrument by designing a component map, ensuring that each component comprises at least 25-30 items, with a balanced representation of positive and negative questions. These items are derived from the document review conducted in the first step.
- 3) An expert review is then conducted, wherein at least three experts in behavioural science evaluate the content validity of the instrument. The questions are subsequently revised and reassessed for face validity to ensure alignment with the objectives, content, theory, and measurement standards (Kemper, 2020; Sireci, 1998).
- 4) Pilot testing follows, wherein the instrument is administered to a sample of 120 individuals who closely resemble the target group. The data collected during this phase are coded for statistical analysis to evaluate the instrument's quality.
- 5) Item quality testing is conducted using an independent-sample t-test with a criterion of 30%, requiring t-values to exceed 2.00 (Sedgwick, 2010), along with Pearson's correlation coefficient analysis, where R-values must be greater than 0.20 (Obilor et al., 2018). Items meeting these criteria should align with the variable definitions outlined in the instrument's map. In cases where insufficient items meet the criteria, priority is given to t-values, as R-values indicate alignment with other items but do not reflect discriminatory power.
- 6) Finally, exploratory factor analysis is performed using principal component analysis (PCA) and Varimax orthogonal rotation to evaluate five key criteria (Hair Jr et al., 2017) the Kaiser-Meyer-Olkin measure of sample adequacy must be at least 0.600, 2) the chi-square statistic must be statistically significant, 3) eigenvalues for components must be at least 1.00, 4) factor loadings must meet a minimum threshold of 0.300, and 5) collectively, all components must account for at least 50% of the variance in the variables.

This facilitates the formulation of the first hypothesis, H1: Exploratory Factor Analysis (EFA) can effectively evaluate the cybersecurity behaviour measurement instrument, requiring a minimum of four items per component, a Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy of no less than 0.600 and a factor loading threshold of at least 0.600, which exceeds the conventional standard. Furthermore, the second hypothesis is articulated as H2: Exploratory Factor Analysis (EFA) can account for more than 60% of the variance in cybersecurity behaviour (Tucker et al., 1997).

- 7) Subsequently, Confirmatory Factor Analysis (CFA) is performed to conduct

second-order factor analysis, thereby assessing construct validity (Smith, 2005). Five criteria are employed to evaluate the fit between the model and the empirical data, referred to as fit measures: 1) Chi-square statistics (McHugh, 2013) Root Mean Square Error of Approximation (Browne et al., 2002) Comparative Fit Index, 4) Tucker-Lewis Index (Cai et al., 2023) Standardized Root Mean Square Residual (Pavlov et al., 2021).

This culminates in the formulation of the third hypothesis, H3: CFA will validate that the model satisfactorily fits the empirical data, as evidenced by the fit indices conforming to the five academic standards outlined previously.

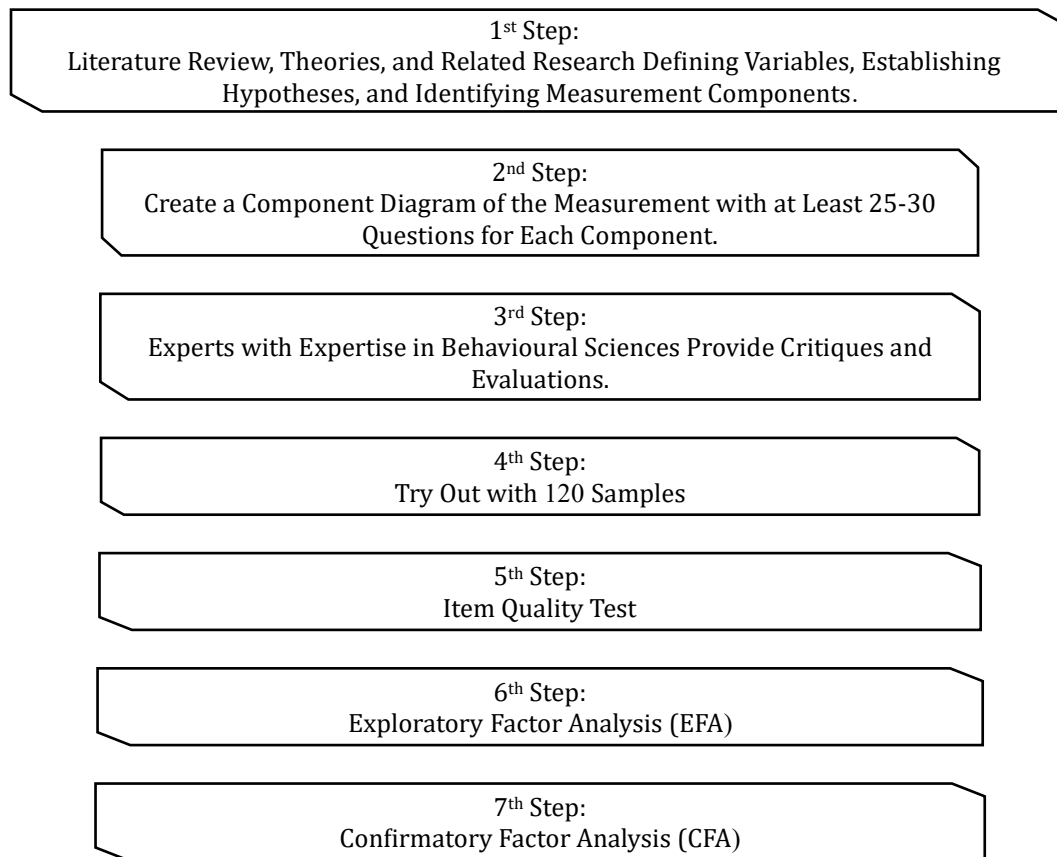


Figure 2: Steps in Developing the Cybersecurity Behaviour Measurement Instrument.

The measurement instrument utilized in this study is a cybersecurity behaviour questionnaire comprising an initial pool of 100 items, developed by the researchers based on core concepts of cybersecurity. This tool encompasses four components, as detailed in Table 2: 1) Cyber Threat Awareness, 2) Cyber Threat Knowledge, 3) Cyber Threat Experience, and 4) Self-Protection from Cyber Threats. Each component features a balanced distribution of positively and negatively phrased statements. Respondents rate each item on a 6-point scale, employing the Summated Rating Method, with options ranging from "Most True" to "Not True at All." The overall reliability of the instrument is calculated to average 0.887 across 67 items. All items

underwent a comprehensive quality assessment following the seven key steps of measurement tool development, as illustrated in Figure 2.

Table 2: Components of the Cybersecurity Behaviour Measurement Instruments.

Cybersecurity Behaviour Measurement				
Code	A	B	C	D
Components	Cybersecurity Threat Awareness	Cybersecurity Behaviour	Cybersecurity Experiences	Cybersecurity Threat Self-Protection
Number of Initial Items	25 Items	25 Items	25 Items	25 Items
Positive Statements	13 Items	12 Items	12 Items	13 Items
Negative Statements	12 Items	13 Items	13 Items	12 Items
IOC Assessment Results	15 Items Remaining	18 Items Remaining	16 Items Remaining	18 Items Remaining
Average Reliability	0.889	0.892	0.884	0.884

Data Analysis

The research undertaken to develop and evaluate the measurement instrument employed three distinct types of statistical analyses. The first type involved statistics for assessing the quality of individual items, specifically utilizing Independent-Sample t-tests (Sedgwick, 2010) and Pearson's Correlation Coefficient Analysis (Obilor et al., 2018). The second type consisted of Factor Analysis aimed at exploring the dimensions or structures of specific characteristics of the items, which included both Exploratory Factor Analysis and Confirmatory Factor Analysis. The third type involved Inferential Statistics, with a particular focus on SEM Analysis (Stein et al., 2012).

Results

From the Index of Objective Congruence (IOC) assessment conducted by a panel of behavioural science experts, a total of 67 questions met the evaluation criteria. These questions underwent preliminary quality testing, where their discriminating power was evaluated using the Independent-Sample t-test (Sedgwick, 2010) and their correlation assessed through Pearson's Correlation Coefficient Analysis (Obilor et al., 2018). The analysis confirmed the presence of all four components, with 21 items meeting the criteria, thus supporting Hypothesis 1. The average reliability score was 0.885 (Table 3). The components that passed the criteria included: 1) Cybersecurity Threat Awareness (4 items), 2) Cybersecurity Threat Knowledge (6 items), 3) Cybersecurity Threat Experience (5 items), and 4) Cybersecurity Threat Self-Protection (6 items).

Table 3: Preliminary Quality Assessment of the Cybersecurity Behaviour Measurement Instruments

Number	Code	Inferential Statistics: Parametric Statistics			
		T-test	R	Cronbach's Alpha	Communalities
1 st Component: Cybersecurity Threat Awareness					
1.	A14	3.708	0.264	0.890	0.673
2.	A7	2.661	0.272	0.887	0.617
3.	A11	2.190	0.240	0.890	0.649
4.	A1	2.969	0.285	0.889	0.514
2 nd Component: Cybersecurity Threat Knowledge					
1.	B14	2.850	0.281	0.887	0.704
2.	B11	3.259	0.349	0.890	0.652
3.	B7	5.236	0.492	0.888	0.590
4.	B13	3.199	0.321	0.893	0.664
5.	B5	2.703	0.327	0.892	0.779
6.	B9	3.242	0.289	0.894	0.562
3 rd Component: Cybersecurity Threat Experience					
1.	C14	4.779	0.471	0.882	0.739
2.	C11	3.655	0.383	0.881	0.724
3.	C3	6.076	0.514	0.882	0.675
4.	C9	2.946	0.478	0.882	0.666
5.	C7	5.663	0.496	0.884	0.755
4 th Component: Cybersecurity Threat Self-Protection					
1.	D12	5.602	0.577	0.876	0.816
2.	D13	3.477	0.442	0.880	0.643
3.	D10	4.289	0.503	0.877	0.588
4.	D15	4.765	0.525	0.884	0.634
5.	D6	3.346	0.493	0.883	0.649
6.	D4	3.653	0.455	0.884	0.539

Note: This Research Places Greater Emphasis on the T-Value than the R-Value, with the Selection Criteria Being a T-Value ≥ 2.00 and an R-Value ≥ 0.20 .

The EFA using Principal Component Analysis and Varimax Orthogonal Rotation identified 21 items meeting the criteria, with factor loadings ranging from 0.608 to 0.871 and Eigenvalues exceeding 1 (Hair Jr et al., 2017)(Table 4). Communalities ranged from 0.514 to 0.816, and all factor loadings were statistically significant at the 0.05 level. The analysis explained 64.187% of the

variance in cybersecurity behaviour among undergraduate students, thus supporting Hypothesis 2. The Kaiser-Meyer-Olkin Measure of Sampling Adequacy was 0.894, surpassing the minimum criterion of 0.600, while Bartlett's Test of Sphericity yielded a value of 6211.453, indicating strong correlations among the 21 items.

Table 4: Cumulative Percentage and Factor Loading of the Cybersecurity Behaviour Measurement Instruments

Code	Questions That Pass the Standard Criteria	Anti-Image	Factor Loading F1	F2	F3
1 st Component: Cybersecurity Threat Awareness					
A14+	Every time I use a banking application to transfer money, I make sure to log out once I finish using it.	0.287		0.793	
A7+	I usually avoid connecting to public wireless networks (Wi-Fi) because I believe that my personal information could be accessed by others.	0.399		0.775	
A11+	I often think that disclosing personal identification on social networks can create security risks for individuals.	0.352		0.743	
A1+	I always set complex passwords that are difficult for others to guess when accessing my information.	0.491		0.655	
2 nd Component: Cybersecurity Threat Knowledge					
B14+	I usually avoid opening or downloading files from unknown senders.	0.323		0.821	
B11+	I try to avoid using public computers, but if necessary, I ensure not to save any personal information.	0.352		0.783	
B7-	I often set my phone to automatically connect to public wireless networks (Wi-Fi) because I prioritize convenience.	0.406		0.759	
B13+	I always read the privacy policy before confirming.	0.319		0.705	
B5-	I tend to write down my passwords in places where they are easy for me to find.	0.275		0.701	
B9+	I never save passwords or allow my information to be remembered on devices in public spaces.	0.418		0.611	
3 rd Component: Cybersecurity Threat Experience					
C14-	I have experienced or know someone close who has been deceived by an individual impersonating someone else or using a fake profile.	0.305			0.830
C11-	I have accessed or filled in personal information on a website that was created to scam and steal my personal data.	0.292			0.780
C3-	I have disabled antivirus software on my computer to download files from a website.	0.344			0.748
C9-	I think it is difficult to differentiate between legitimate websites and fake ones.	0.316			0.741
C7-	I have been attacked by a computer virus because I wanted to watch movies or listen to music that were available for free download.	0.260			0.717

4 th Component: Cybersecurity Threat Self-Protection			
D12+	I will not post pictures or anything personal in public spaces because there is a risk that the information may be used for malicious purposes.	0.372	0.871
D13+	I will set my privacy settings to avoid receiving spam or advertising messages.	0.398	0.787
D10+	I do not share personal information with others unless I know their true purpose.	0.364	0.755
D15+	Unless necessary, I usually do not use other people's communication devices and do not let others use my devices either.	0.436	0.724
D6 +	I believe that installing tools or devices to prevent cyber-attacks helps protect me from cyber dangers.	0.358	0.661
D4 +	I do not share my current location that can identify my identity on social media.	0.392	0.608
Initial Eigenvalues		10.110	5.336 2.526
% of Variance		36.107	19.058 9.022
Cumulative %		36.107	55.16564.187
Kasier-Meyer-Olkin Measure of Sampling Adequacy		0.894	
Bartlett's Test of Sphericity		6211.453	
df		51	

The results of the Confirmatory Factor Analysis indicate a good fit of the cybersecurity behaviour measurement model with the empirical data, thereby supporting Hypothesis 3, as the indices meet the standard criteria (Table 5 and Figure 3). The highest coherence was observed in Path D of the self-protection component ($\beta = 0.904$, $R^2 = 0.812$), followed by Path C of the experience component ($\beta = 0.865$, $R^2 = 0.857$), and Path B of the knowledge component ($\beta = 0.824$, $R^2 = 0.795$). Path A of the awareness component had the least influence ($\beta = 0.725$, $R^2 = 0.791$).

Table 5: Indices of Model Fit for the Cybersecurity Behaviour Measurement Instruments

Statistics	Criteria for Consideration	Statistics in the Model (Total Group)	Result of Consideration
Chi-Square Value	Not Statistically Significant	94.392	Accepted
Degrees of Freedom	Not Statistically Significant	59	Accepted
P-Value	Not Statistically Significant	0.531	Accepted
Root Mean Square Error of Approximation	≤ 0.06	0.044	Accepted
Comparative Fit Index	≥ 0.95	0.987	Accepted
Tucker – Lewis Index	≥ 0.95	0.986	Accepted
Standardized Root Mean Square	≤ 0.08	0.062	Accepted

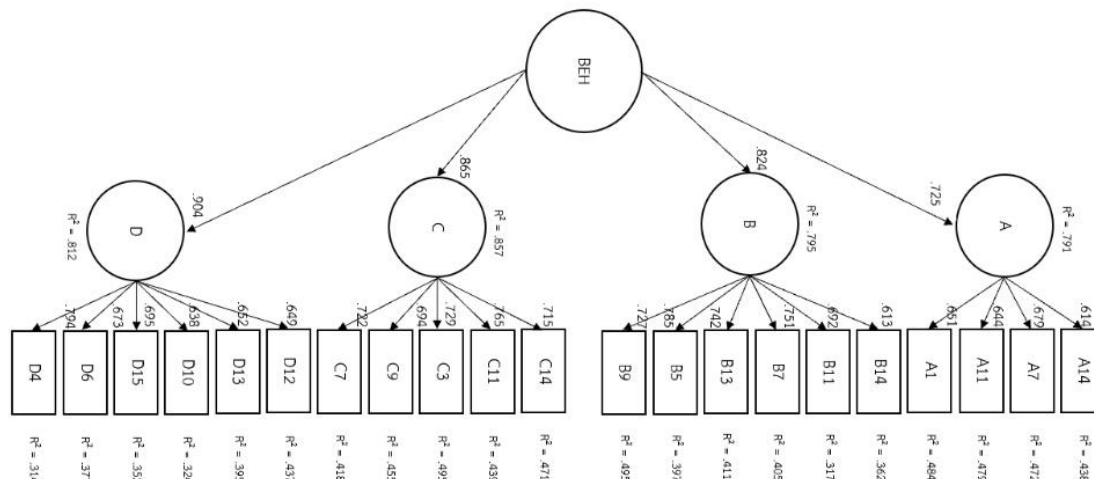


Figure 3: Fit of the Cybersecurity Behaviour Measurement Instrument Model with Empirical Data.

Discussion and Conclusion

This research identified 21 items that met the evaluation criteria across all four components, yielding an average reliability coefficient of 0.885. The measurement tool effectively explained 64.187% of the cybersecurity behaviour exhibited by undergraduate students. These findings are consistent with prior studies (Bidgoli et al., 2016; Djeki et al., 2024), which serve as a framework for research focused on developing and evaluating measurement instruments that incorporate the second level of Maslow's Hierarchy of Needs alongside cybersecurity concepts. Additionally, the results align with earlier investigations into cybersecurity issues among undergraduate students (Ahmad et al., 2021; Yan et al., 2018), thereby supporting all three hypotheses.

In examining each hypothesis, it was observed that the EFA required an adjustment in the factor loading threshold from 0.300 to 0.600. This adjustment enhanced the academic rigor of the items that met the criteria but potentially led to a reduction in the number of qualifying items. Consequently, Component 1, focusing on awareness of cybersecurity threats, was limited to four items, still satisfying the hypothesis requirement of a minimum of four questions. The Kaiser-Meyer-Olkin Measure of Sampling Adequacy remained within acceptable limits. Furthermore, CFA indicated a strong fit of the structural model, with all five indices meeting the established criteria, thus supporting Hypothesis 3 across all dimensions and demonstrating the clarity of the measurement instrument's components. The analysis conducted using this model constitutes a second-order factor analysis and provides robust evidence for construct validity (Smith, 2005).

Limitations

The cybersecurity behaviour measurement instrument developed in this study is specifically applicable to undergraduate students. To enhance its applicability, future adaptations should target other populations, such as school students or children. This approach would facilitate comparative analyses across diverse demographic groups,

potentially uncovering significant variations in cybersecurity behaviours. Moreover, considering the dynamic landscape of cyber threats, it is imperative that the instrument's questions undergo regular updates to ensure their relevance to contemporary societal contexts.

Future Research Directions

For future research, it is advisable that the measurement instrument undergo an additional round of CFA to statistically validate any potential differences. The instrument could also be employed to investigate causal factors related to cybersecurity behaviour among undergraduate students. This may involve conducting causal-comparative research or examining cause-and-effect relationships by focusing on various independent or dependent variables to elucidate differences between groups. Furthermore, the tool could be adapted to assess students' cybersecurity readiness, potentially identifying at-risk groups who would benefit from targeted training. Additionally, the instrument could be refined to function as a social index, thereby raising awareness and guiding proactive prevention strategies, as cybersecurity continues to be a dynamic and critical issue. In this study, the researchers have already integrated the questionnaire into a mobile application that enables undergraduate students to self-assess and evaluate their understanding of cybersecurity issues. A handbook on cybersecurity knowledge has also been developed for interested individuals. Moreover, the findings from this research have been presented for further policy discussions, potentially contributing to strategic development plans.

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